

Experiment Tables

Tensor Greedy Double Block Extended Kaczmarz (TGDBEK)
Proposed method marked * / bold

Example 4.1 — Dense overdetermined tensor systems

$\mathcal{A} \in \mathbb{R}^{500 \times n \times 10}$, $\mathcal{B} \in \mathbb{R}^{500 \times 10 \times 10}$, noise = 10^{-3} , max iter = 800, $\eta = 0.6$, $\tau = 10$, 5 trials.

Table 1: Number of iterations (IT) for dense overdetermined tensor systems $\mathcal{A} \in \mathbb{R}^{500 \times n \times 10}$, averaged over 5 independent trials. Bold indicates the proposed TGDBEK method. “—” denotes no convergence within the iteration budget.

Method	$n = 20$	$n = 30$	$n = 40$	$n = 50$	$n = 60$	$n = 80$
TREK	542	—	—	—	—	—
TREBK	36	66	95	126	167	259
TREGBK	36	62	89	117	148	217
TGDBEK	18	30	28	36	33	48

Table 2: CPU time (seconds) for dense overdetermined tensor systems, averaged over 5 independent trials.

Method	$n = 20$	$n = 30$	$n = 40$	$n = 50$	$n = 60$	$n = 80$
TREK	2.225	3.244	2.947	2.952	2.788	2.855
TREBK	1.003	1.632	2.153	2.897	3.691	6.118
TREGBK	1.051	1.678	2.368	3.162	3.916	6.451
TGDBEK	0.859	1.303	1.285	1.708	1.732	3.100

Example 4.2 — Sparse tensor systems (SuiteSparse)

Tolerance 10^{-6} , $\tau = 10$, η per matrix.

Examples 4.3–4.5 — Image deblurring

Gaussian blur ($\sigma = 4$), noise added, tolerance 10^{-5} , $\eta = 0.5$, $\tau = 10$.

Example 4.6 — Effect of η on TGDBEK

Table 3: Results on sparse tensor systems from SuiteSparse. IT: iterations; CPU: seconds; RSE: relative squared error $\|\mathcal{X}^{(k)} - \mathcal{X}^*\|_F^2 / \|\mathcal{X}^*\|_F^2$. “—” denotes no convergence. * marks the proposed method.

Matrix	Method	IT	CPU (s)	RSE
nos5	TREK	300	1.239	9.288×10^{-2}
	TREBK	138	4.054	9.888×10^{-7}
	TREGBK	106	3.399	9.559×10^{-7}
	*TGDBEK	44	2.828	9.659×10^{-7}
ash85	TREK	300	1.173	1.386×10^{-1}
	TREBK	171	2.544	9.463×10^{-7}
	TREGBK	300	4.401	1.028×10^{-6}
	*TGDBEK	128	2.964	9.716×10^{-7}
Cities	TREK	6000	21.075	3.807×10^{-3}
	TREBK	1916	15.799	9.815×10^{-7}
	TREGBK	5740	46.848	9.881×10^{-7}
	*TGDBEK	1107	15.355	9.783×10^{-7}
WorldCities	TREK	6000	22.366	—
	TREBK	3	0.155	7.770×10^{-7}
	TREGBK	228	5.413	8.757×10^{-7}
	*TGDBEK	1	0.070	5.527×10^{-7}

Table 4: Color image deblurring (flower, $200 \times 200 \times 3$, noise = 10^{-2} , max iter = 1000).

Method	IT	CPU (s)	RSE	PSNR (dB)	SSIM
TREK	1000	3.580	7.885×10^{-1}	13.04	0.2654
TREBK	1000	9.413	1.428×10^{-2}	30.19	0.8405
TREGBK	1000	16.637	1.172×10^{-3}	30.32	0.8432
*TGDBEK	480	14.441	9.877×10^{-6}	30.32	0.8432

Table 5: Color image deblurring (portrait, $200 \times 200 \times 3$, noise = 10^{-1} , max iter = 3000).

Method	IT	CPU (s)	RSE	PSNR (dB)	SSIM
TREK	3000	9.920	7.175×10^{-1}	12.52	0.1786
TREBK	2577	24.241	9.036×10^{-6}	12.13	0.1582
TREGBK	1714	24.862	9.785×10^{-6}	12.13	0.1582
*TGDBEK	541	17.832	9.993×10^{-6}	12.13	0.1582

Table 6: Gray MRI image deblurring (Shepp-Logan, $128 \times 128 \times 27$, noise = 10^{-3} , max iter = 1000).

Method	IT	CPU (s)	RSE	PSNR (dB)	SSIM
TREK	1000	4.536	7.161×10^{-1}	17.75	0.5822
TREBK	1000	60.832	5.407×10^{-4}	58.02	0.9976
TREGBK	1000	69.298	2.437×10^{-5}	58.07	0.9976
*TGDBEK	477	42.851	9.878×10^{-6}	58.07	0.9976

Table 7: Effect of the greedy threshold η on TGDBEK. Dense system $\mathcal{A} \in \mathbb{R}^{500 \times 20 \times 10}$, noise = 10^{-3} .

η	IT	CPU (s)
0.1	1	0.065
0.2	1	0.032
0.3	1	0.032
0.4	1	0.043
0.5	6	0.655
0.6	18	0.966
0.7	37	1.457
0.8	57	1.760
0.9	88	2.353
1.0	362	7.962